Assignment 6: Data Warehousing

Course: Data Acquisition and Management
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1. Executive Summary

In this assignment, we built a data warehouse named **bill_dwh** in PostgreSQL using a **star schema**. The process involved:

1. Creating a dedicated directory for the project.
2. Writing a Bash script (assignment.sh) to automate database creation, script downloads, and data loading.
3. Executing SQL scripts for creating the star schema, loading dimension tables, loading the fact table, and verifying data.
4. Checking results using both command-line logs and **pgAdmin**.

This report documents each step with accompanying screenshots to illustrate the setup, execution, and final verification of the data warehouse.

2. Environment Setup

- **Files & Scripts**:
    - assignment.sh – Main Bash script
    - star-schema.sql – Creates the database tables
    - DimCustomer.sql / DimMonth.sql – Populate dimension tables
    - FactBilling.sql – Populate the fact table
    - verify.sql – Check row counts / data validation
    - assignment.log – Log of script output

3. Directory Structure

All files are kept in ~/Assignment6:

/home/guru/Assignment6

├── assignment.sh

├── billing-datawarehouse.tgz

├── star-schema.sql

├── DimCustomer.sql

├── DimMonth.sql

├── FactBilling.sql

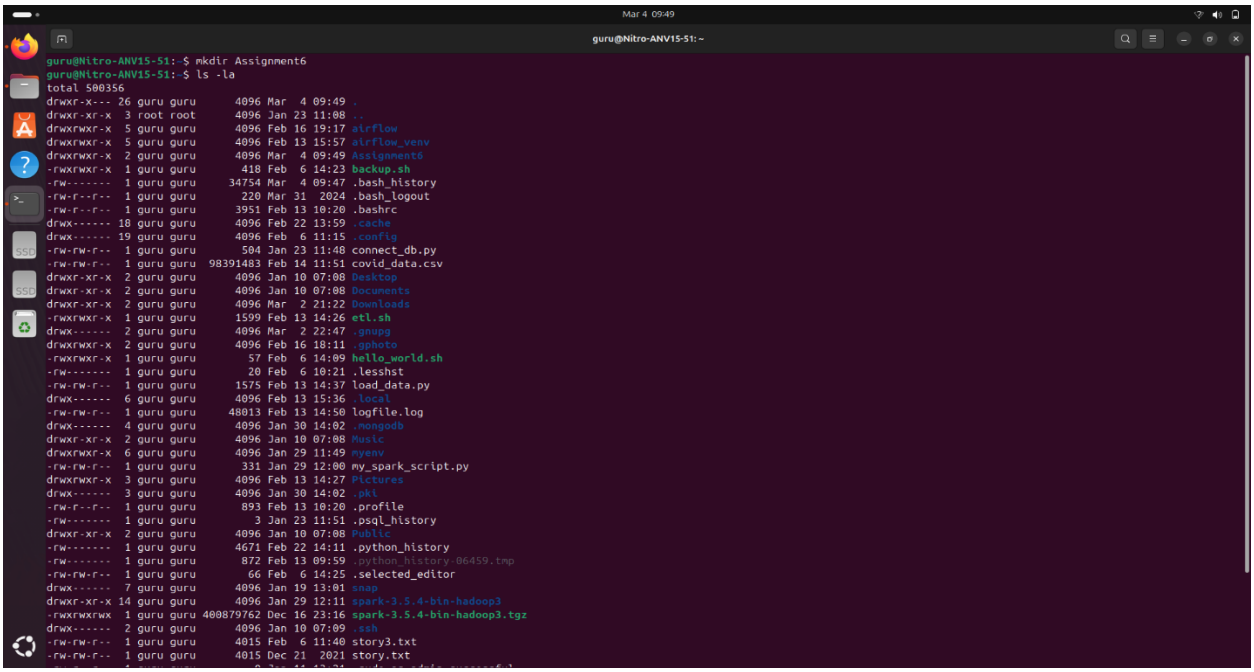
├── verify.sql

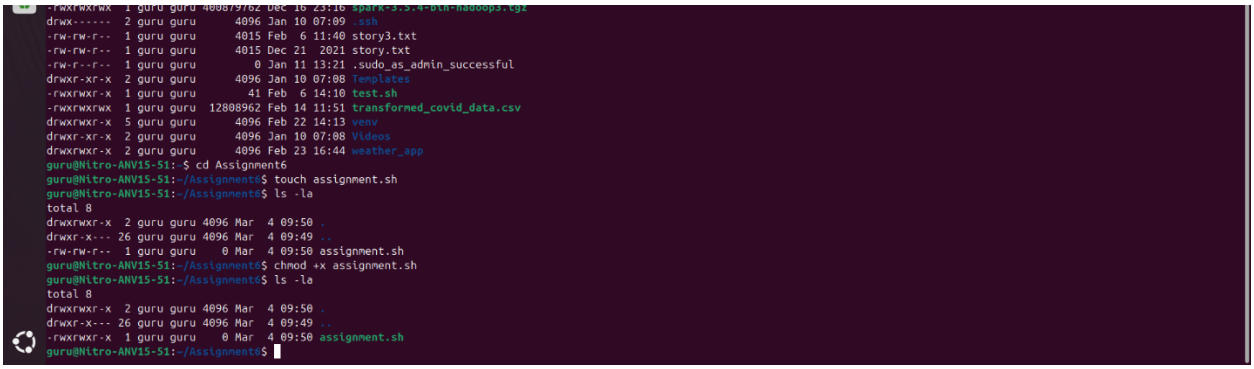
└── assignment.log

4. Step-by-Step Implementation

4.1. Creating the Directory & Script Setup

1. We began by creating a directory Assignment6, then created a Bash script assignment.sh to automate the data warehouse steps.
    - **Command**:
    - mkdir Assignment6
    - cd Assignment6
    - touch assignment.sh
    - chmod +x assignment.sh

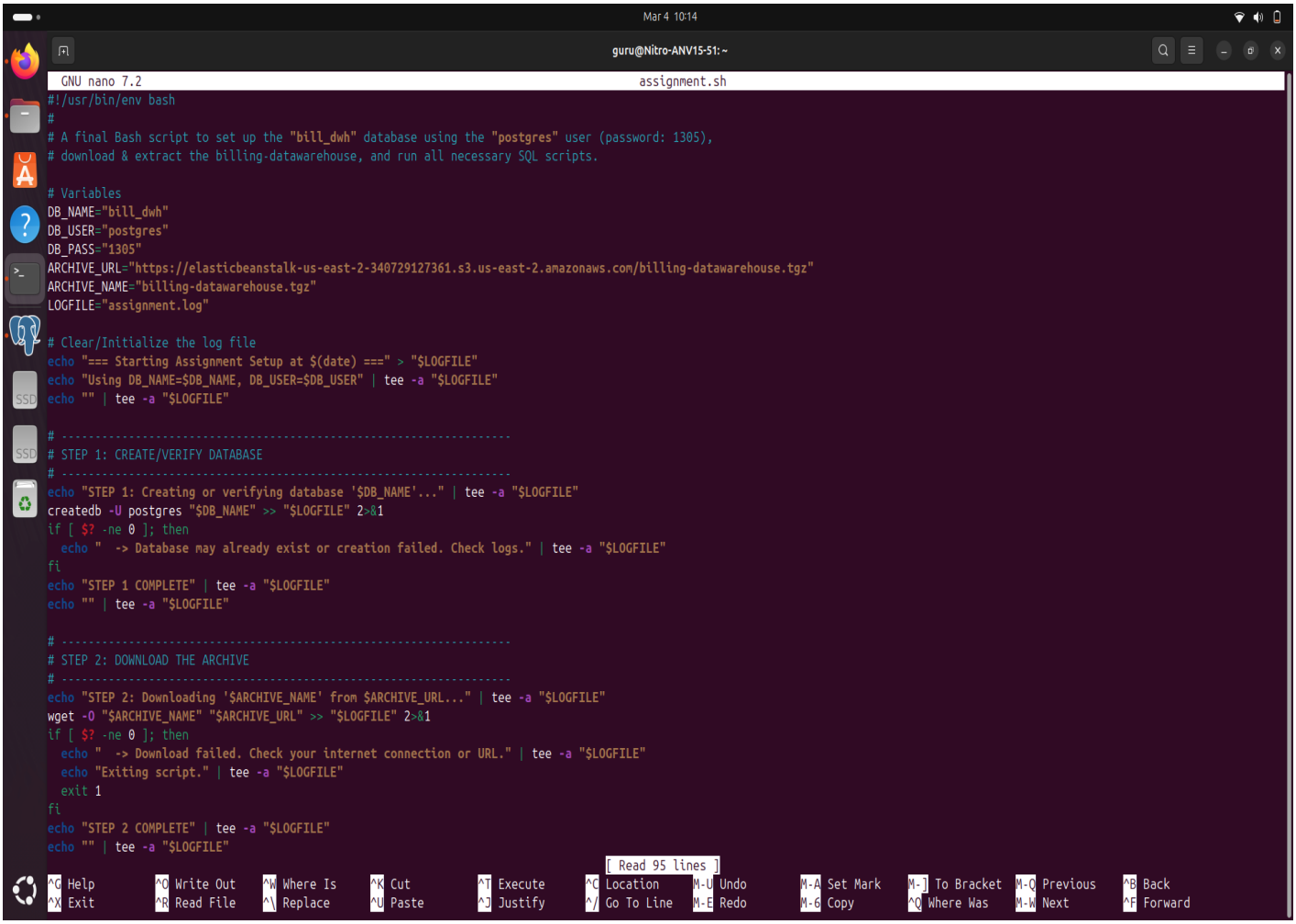


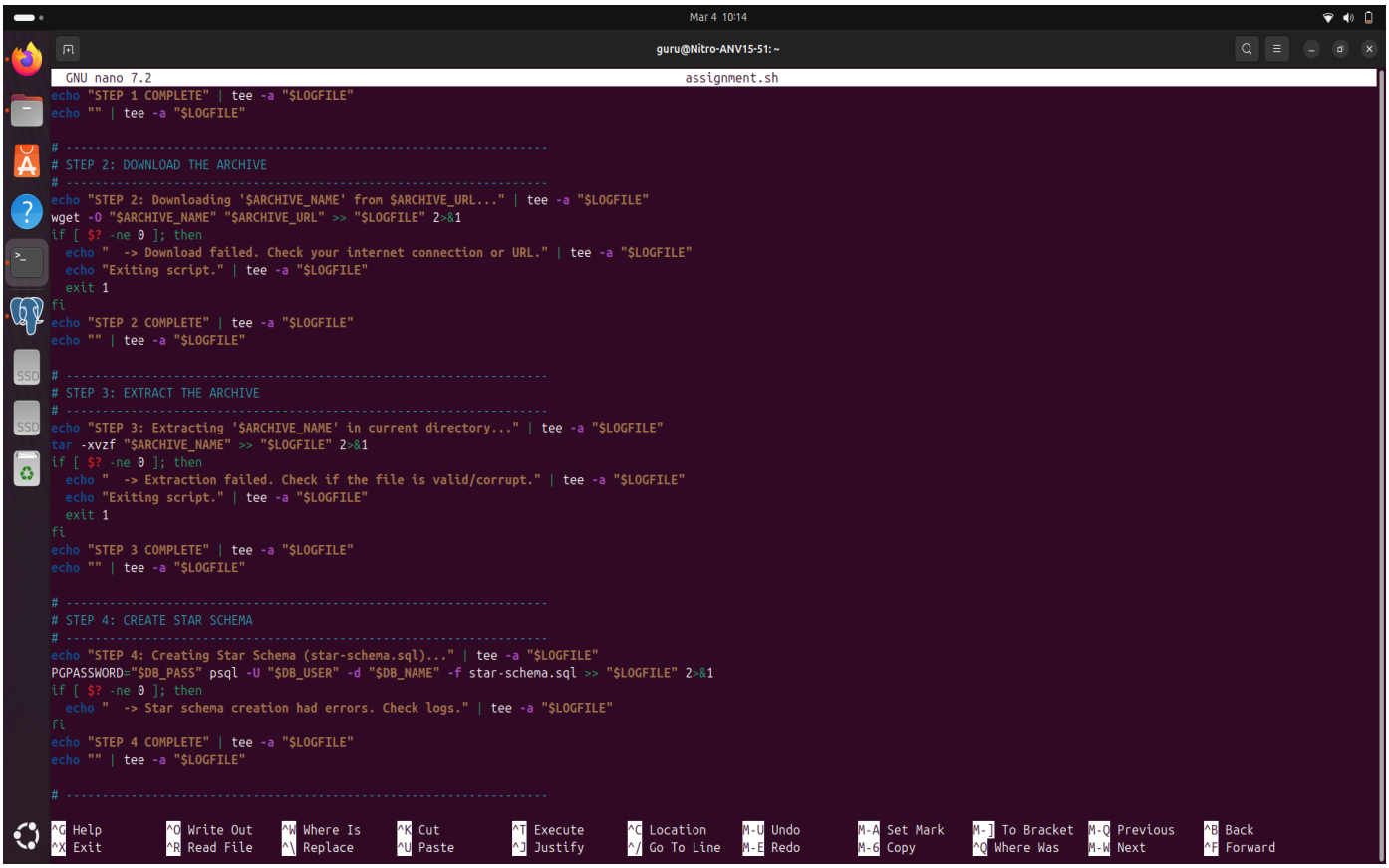


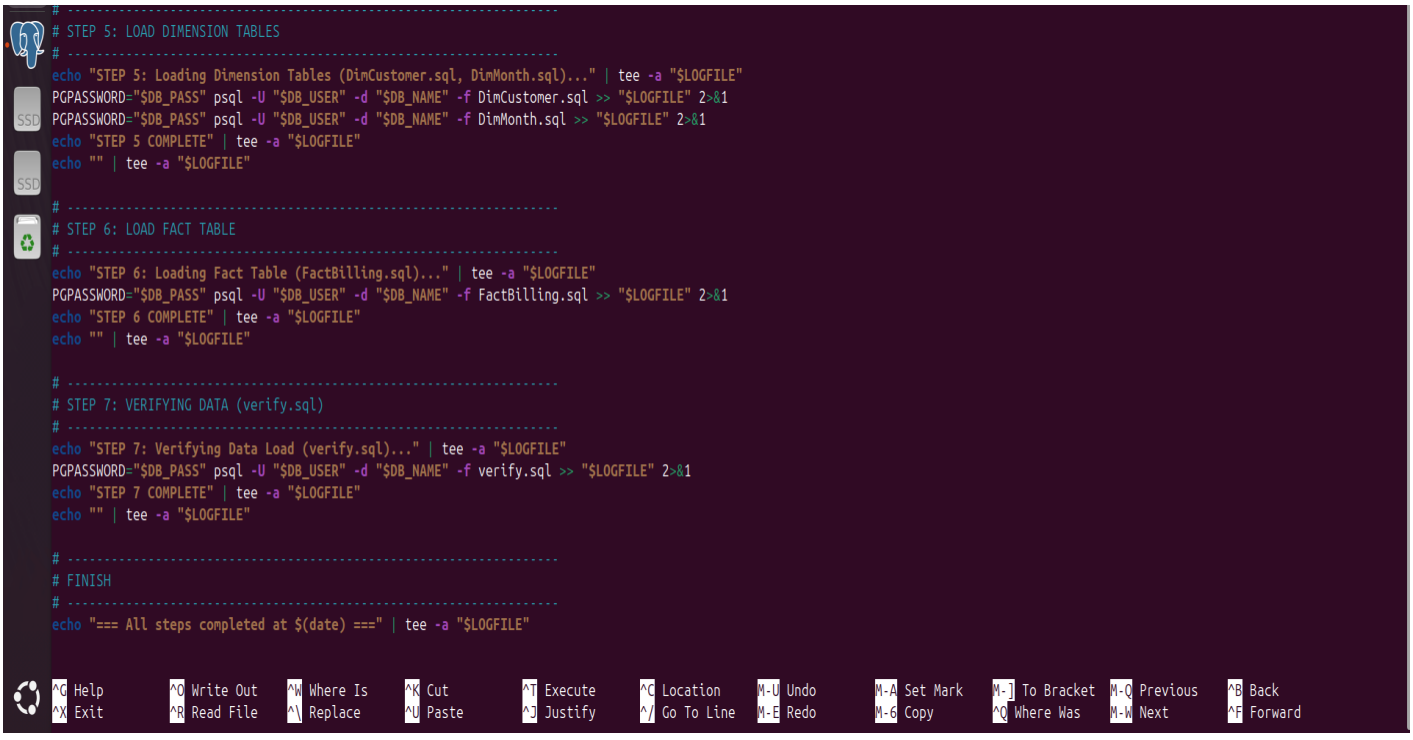
4.2. Editing assignment.sh

The script contains variables for database name, user, and password, plus commands to:

- Create (or verify) the database.
- Download billing-datawarehouse.tgz.
- Extract SQL scripts.
- Execute the star schema creation.
- Load dimension and fact tables.
- Run verify.sql for final checks.
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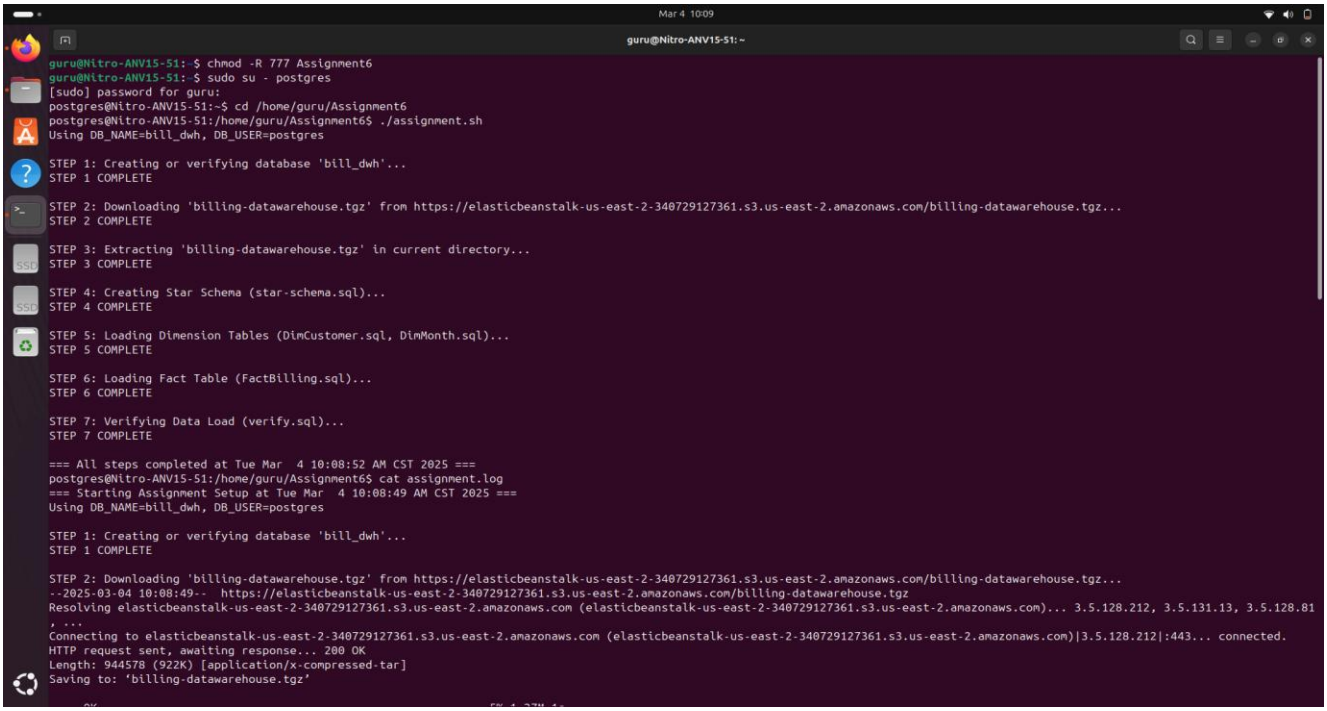



Mar 4 10:14
guru@Nitro-ANV15-51: ~
GNU nano 7.2 assignment.sh
echo "STEP 1 COMPLETE" | tee -a "$LOGFILE"
echo "" | tee -a "$LOGFILE"
--
STEP 2: DOWNLOAD THE ARCHIVE
--
echo "STEP 2: Downloading '$ARCHIVE_NAME' from $ARCHIVE_URL..." | tee -a "$LOGFILE"
wget -O "$ARCHIVE_NAME" "$ARCHIVE_URL" >> "$LOGFILE" 2>&1
if [$? -ne 0]; then
echo " -> Download failed. Check your internet connection or URL." | tee -a "$LOGFILE"
echo "Exiting script." | tee -a "$LOGFILE"
exit 1
fi
echo "STEP 2 COMPLETE" | tee -a "$LOGFILE"
echo "" | tee -a "$LOGFILE"
--
STEP 3: EXTRACT THE ARCHIVE
--
echo "STEP 3: Extracting '$ARCHIVE_NAME' in current directory..." | tee -a "$LOGFILE"
tar -xvzf "$ARCHIVE_NAME" >> "$LOGFILE" 2>&1
if [$? -ne 0]; then
echo " -> Extraction failed. Check if the file is valid/corrupt." | tee -a "$LOGFILE"
echo "Exiting script." | tee -a "$LOGFILE"
exit 1
fi
echo "STEP 3 COMPLETE" | tee -a "$LOGFILE"
echo "" | tee -a "$LOGFILE"
--
STEP 4: CREATE STAR SCHEMA
--
echo "STEP 4: Creating Star Schema (star-schema.sql)..." | tee -a "$LOGFILE"
PGPASSWORD="$DB_PASS" psql -U "$DB_USER" -d "$DB_NAME" -f star-schema.sql >> "$LOGFILE" 2>&1
if [$? -ne 0]; then
echo " -> Star schema creation had errors. Check logs." | tee -a "$LOGFILE"
fi
echo "STEP 4 COMPLETE" | tee -a "$LOGFILE"
echo "" | tee -a "$LOGFILE"
--
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket M-Q Previous ^B Back
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was M-W Next ^F Forward


--
STEP 5: LOAD DIMENSION TABLES
--
echo "STEP 5: Loading Dimension Tables (DimCustomer.sql, DimMonth.sql)..." | tee -a "$LOGFILE"
PGPASSWORD="$DB_PASS" psql -U "$DB_USER" -d "$DB_NAME" -f DimCustomer.sql >> "$LOGFILE" 2>&1
PGPASSWORD="$DB_PASS" psql -U "$DB_USER" -d "$DB_NAME" -f DimMonth.sql >> "$LOGFILE" 2>&1
echo "STEP 5 COMPLETE" | tee -a "$LOGFILE"
echo "" | tee -a "$LOGFILE"
--
STEP 6: LOAD FACT TABLE
--
echo "STEP 6: Loading Fact Table (FactBilling.sql)..." | tee -a "$LOGFILE"
PGPASSWORD="$DB_PASS" psql -U "$DB_USER" -d "$DB_NAME" -f FactBilling.sql >> "$LOGFILE" 2>&1
echo "STEP 6 COMPLETE" | tee -a "$LOGFILE"
echo "" | tee -a "$LOGFILE"
--
STEP 7: VERIFYING DATA (verify.sql)
--
echo "STEP 7: Verifying Data Load (verify.sql)..." | tee -a "$LOGFILE"
PGPASSWORD="$DB_PASS" psql -U "$DB_USER" -d "$DB_NAME" -f verify.sql >> "$LOGFILE" 2>&1
echo "STEP 7 COMPLETE" | tee -a "$LOGFILE"
echo "" | tee -a "$LOGFILE"
--
FINISH
--
echo "=== All steps completed at $(date) ===" | tee -a "$LOGFILE"
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket M-Q Previous ^B Back
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was M-W Next ^F Forward

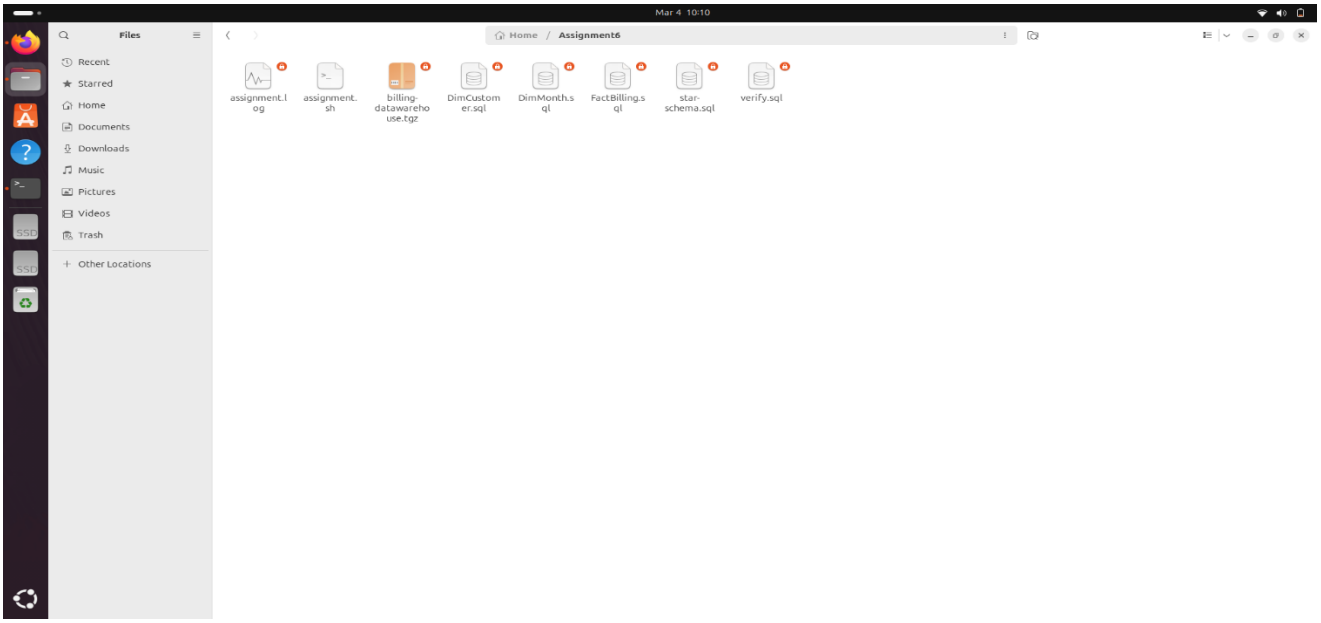
4.3. Running assignment.sh and Verifying Output

Once permissions were resolved (ensuring the postgres user could write logs), we ran the script **assignment.sh**



4.4. Final Directory Listing

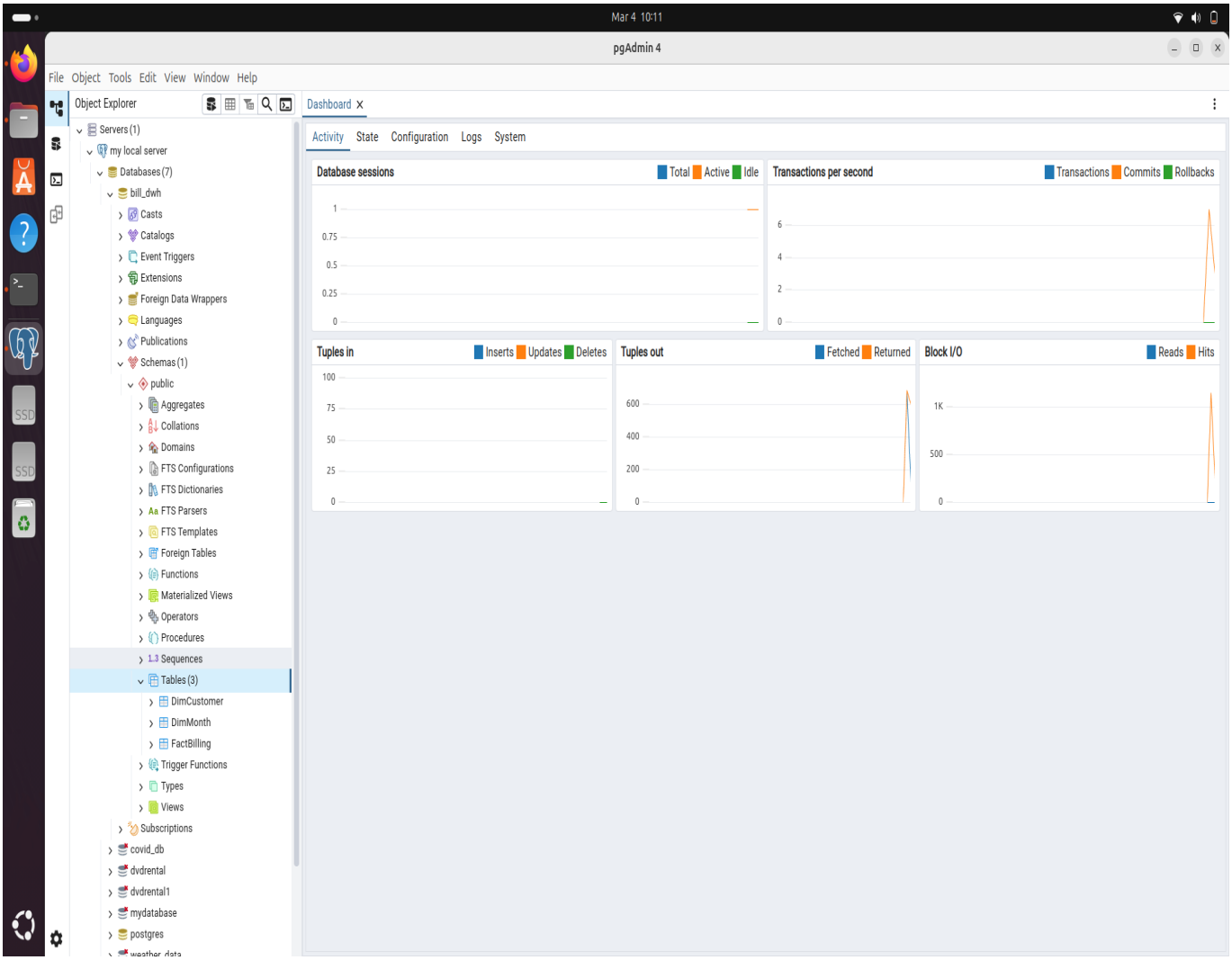
After successful execution:

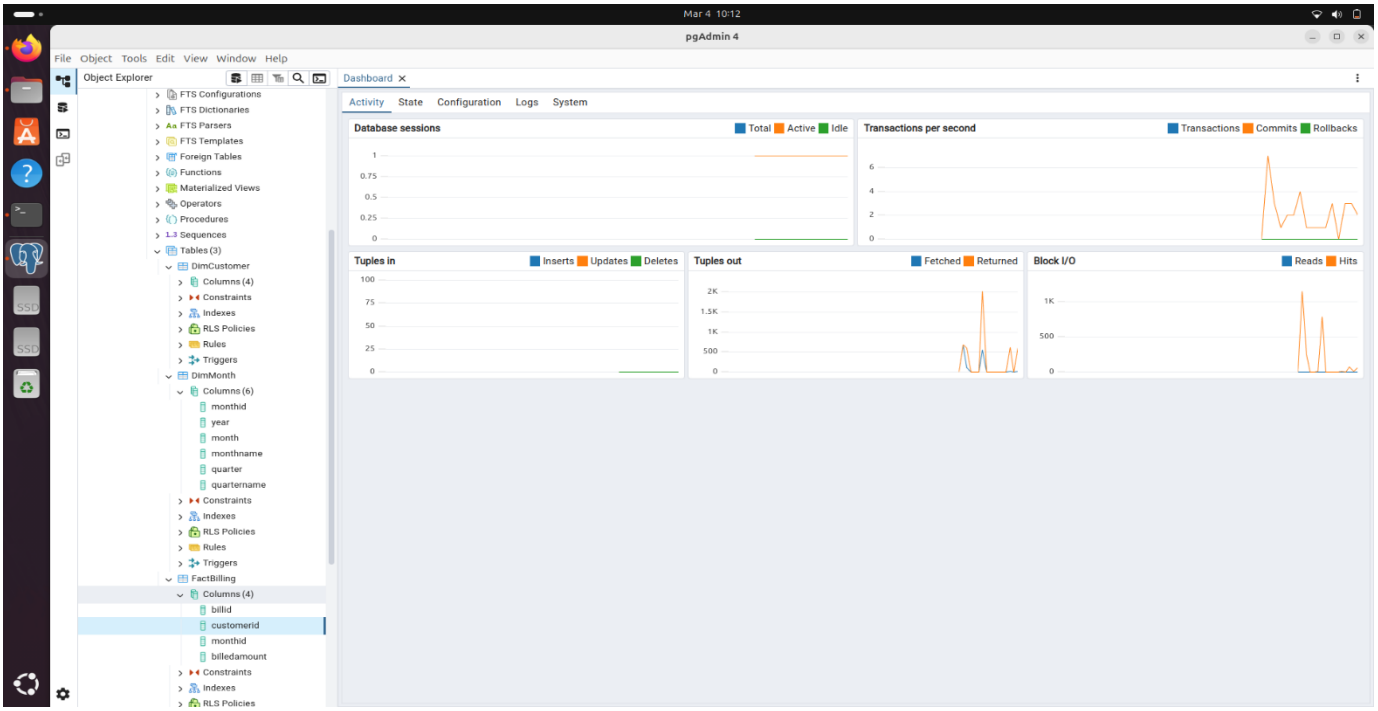


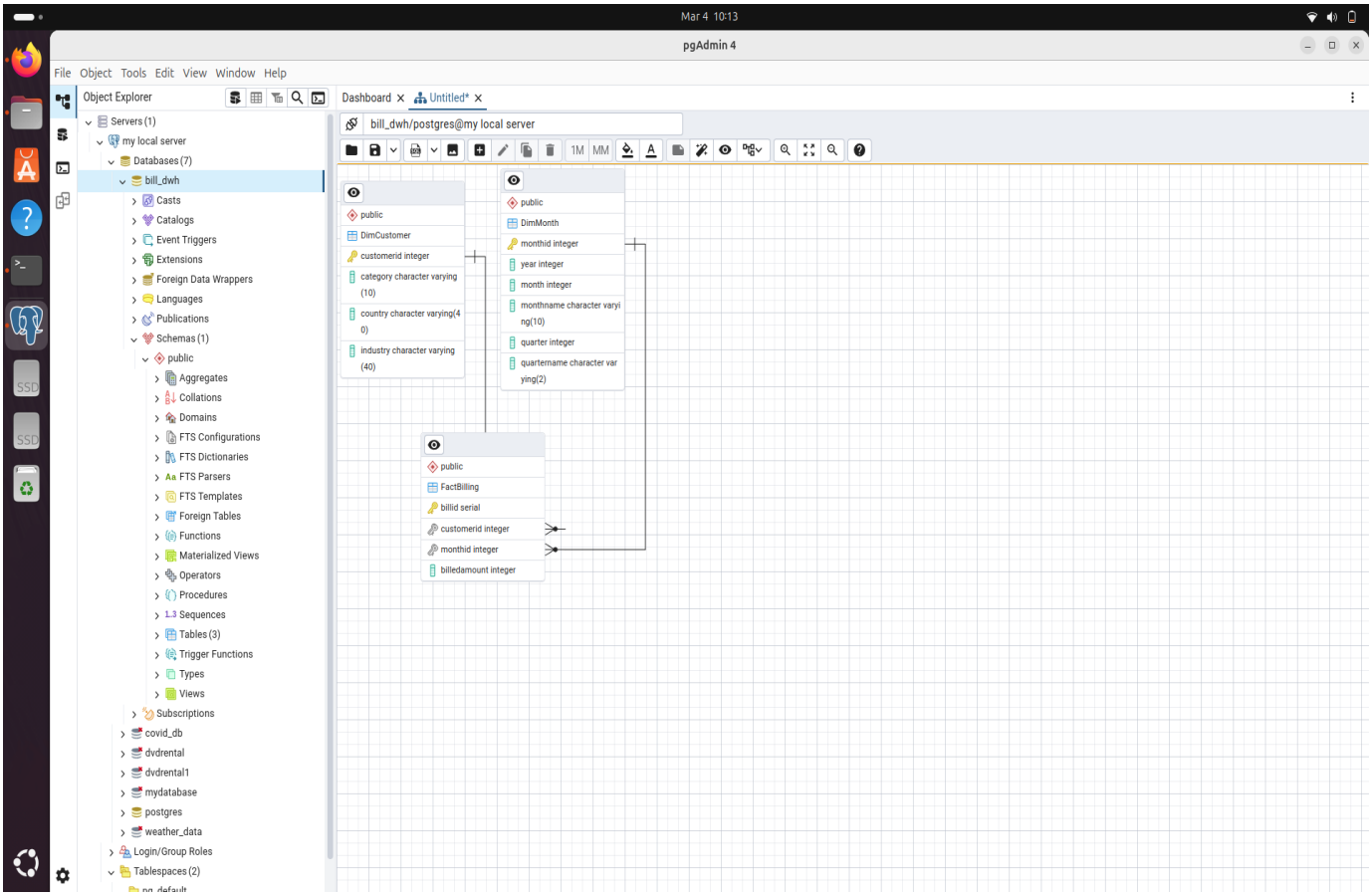
4.5. Checking in pgAdmin

To visualize the final star schema and confirm the data:

1. **Open pgAdmin** and connect to bill_dwh.
2. **Look under** Schemas → public → Tables. We see DimCustomer, DimMonth, FactBilling.
3. **Inspect** the columns and relationships.




Mar 4 10:12
pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
Dashboard
Activity State Configuration Logs System
Database sessions
Total Active Idle
Transactions per second
Transactions Commits Rollbacks
Tuples in
Inserts Updates Deletes
Tuples out
Fetched Returned
Block I/O
Reads Hits
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (3)
DimCustomer
Columns (4)
Constraints
Indexes
RLS Policies
Rules
Triggers
DimMonth
Columns (6)
monthid
year
month
monthname
quarter
quartername
FactBilling
Columns (4)
billid
customerid
monthid
billedamount


Mar 4 10:13
pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
Dashboard
Untitled*
bill_dwh/postgres@my local server
Servers (1)
my local server
Databases (7)
bill_dwh
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (3)
Trigger Functions
Types
Views
Subscriptions
covid_db
dvdrental
dvdrental1
mydatabase
postgres
weather_data
Login/Group Roles
Tablespaces (2)
public
DimCustomer
customerid integer
category character varying (10)
country character varying(40)
industry character varying (40)
public
DimMonth
monthid integer
year integer
month integer
monthname character varying(10)
quarter integer
quartername character varying(2)
public
FactBilling
billid serial
customerid integer
monthid integer
billedamount integer

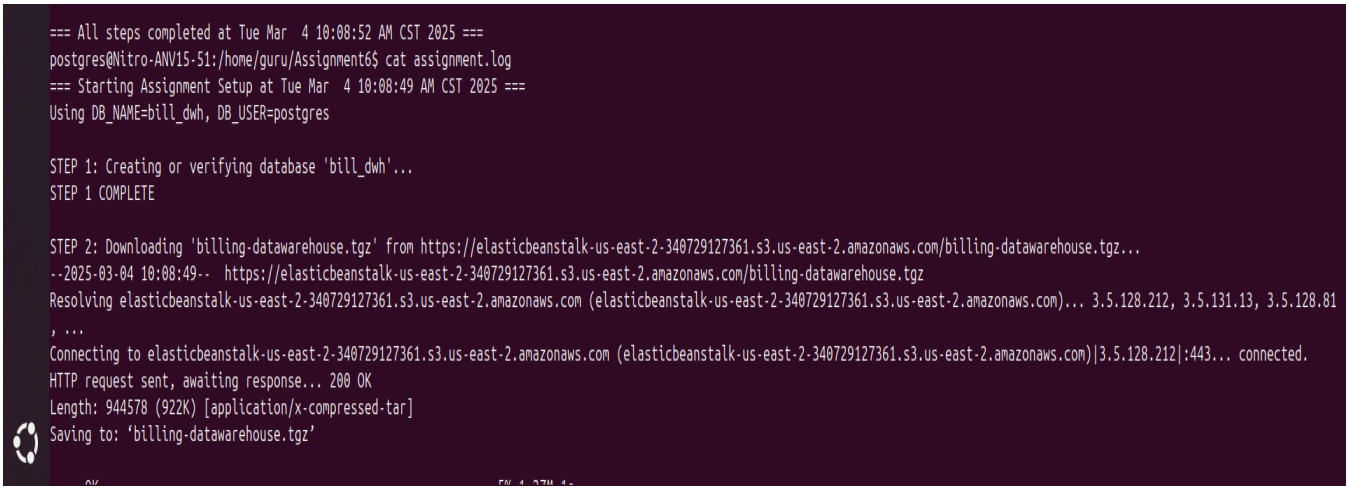
5. Verification and Observations

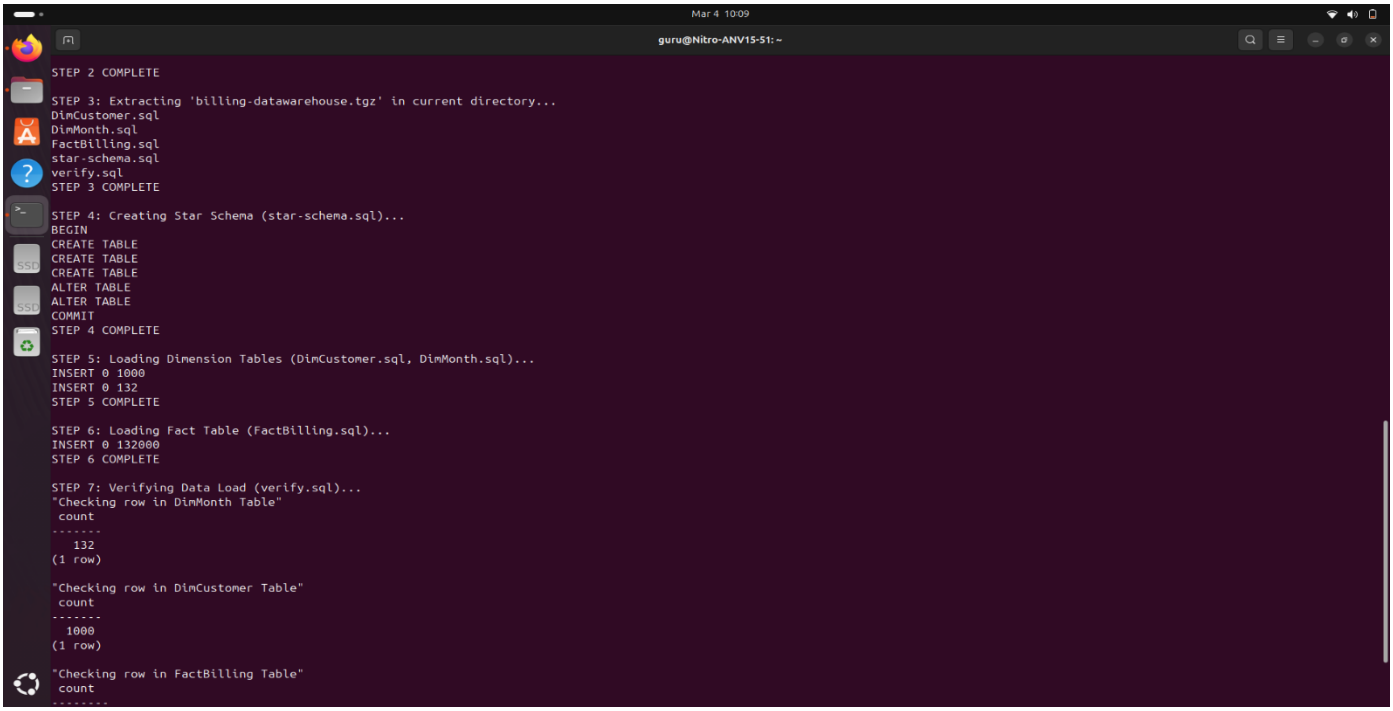
- **Log File**: assignment.log records each command's output and any errors.
- **Row Counts**:
    - DimCustomer: 1000 rows
    - DimMonth: 132 rows
    - FactBilling: 132000 rows

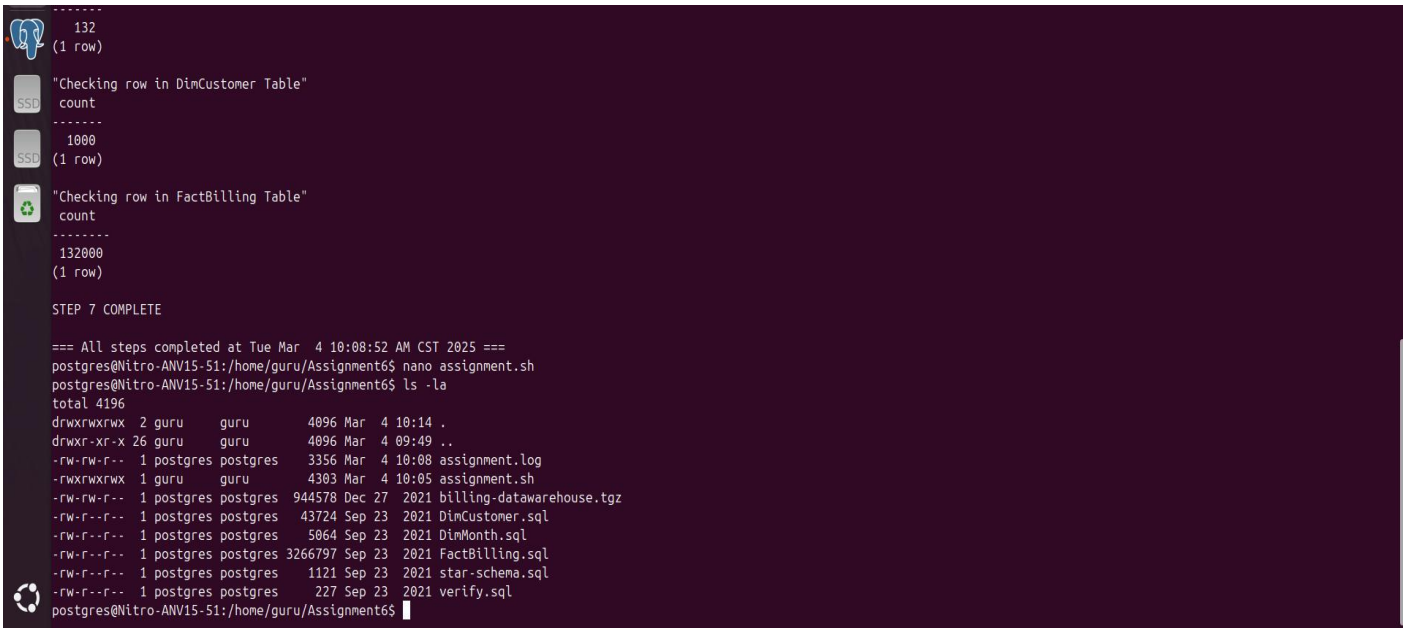
These counts match expectations based on the provided data. The verification script (verify.sql) confirmed the successful loads.

Common Issues Faced:

1. **Peer Authentication**: We resolved "peer authentication failed" by running everything under the OS user postgres and adjusting directory permissions (chmod/chown).
2. **Permission Denied**: We made sure assignment.log was writable by the postgres user or set 777 on Assignment6.
3. **Structure**: The star schema approach with FactBilling referencing dimension keys in DimCustomer and DimMonth.




Mar 4 10:09
guru@Nitro-ANV15-51: ~
STEP 2 COMPLETE
STEP 3: Extracting 'billing-datawarehouse.tgz' in current directory...
DimCustomer.sql
DimMonth.sql
FactBilling.sql
star-schema.sql
verify.sql
STEP 3 COMPLETE
STEP 4: Creating Star Schema (star-schema.sql)...
BEGIN
CREATE TABLE
CREATE TABLE
CREATE TABLE
ALTER TABLE
ALTER TABLE
COMMIT
STEP 4 COMPLETE
STEP 5: Loading Dimension Tables (DimCustomer.sql, DimMonth.sql)...
INSERT 0 1000
INSERT 0 132
STEP 5 COMPLETE
STEP 6: Loading Fact Table (FactBilling.sql)...
INSERT 0 132000
STEP 6 COMPLETE
STEP 7: Verifying Data Load (verify.sql)...
"Checking row in DimMonth Table"
count
132
(1 row)
"Checking row in DimCustomer Table"
count
1000
(1 row)
"Checking row in FactBilling Table"
count


132
(1 row)
"Checking row in DimCustomer Table"
count
1000
(1 row)
"Checking row in FactBilling Table"
count
132000
(1 row)
STEP 7 COMPLETE
=== All steps completed at Tue Mar 4 10:08:52 AM CST 2025 ===
postgres@Nitro-ANV15-51:/home/guru/Assignment6$ nano assignment.sh
postgres@Nitro-ANV15-51:/home/guru/Assignment6$ ls -la
total 4196
drwxrwxrwx 2 guru guru 4096 Mar 4 10:14 .
drwxr-xr-x 26 guru guru 4096 Mar 4 09:49 ..
-rw-rw-r-- 1 postgres postgres 3356 Mar 4 10:08 assignment.log
-rwxrwxrwx 1 guru guru 4303 Mar 4 10:05 assignment.sh
-rw-rw-r-- 1 postgres postgres 944578 Dec 27 2021 billing-datawarehouse.tgz
-rw-r--r-- 1 postgres postgres 43724 Sep 23 2021 DimCustomer.sql
-rw-r--r-- 1 postgres postgres 5064 Sep 23 2021 DimMonth.sql
-rw-r--r-- 1 postgres postgres 3266797 Sep 23 2021 FactBilling.sql
-rw-r--r-- 1 postgres postgres 1121 Sep 23 2021 star-schema.sql
-rw-r--r-- 1 postgres postgres 227 Sep 23 2021 verify.sql
postgres@Nitro-ANV15-51:/home/guru/Assignment6$

6. Conclusion

In this assignment, we:

1. **Created** a new database bill_dwh in PostgreSQL.
2. **Built** a star schema with dimension and fact tables.
3. **Loaded** all data via SQL scripts and verified row counts.
4. **Examined** the final structure in pgAdmin to confirm correctness.

Through this exercise, we gained hands-on experience in:

- Automating data warehouse setup with a Bash script.
- Handling PostgreSQL authentication (peer vs. password).
- Validating data integrity with simple verification queries.
- Understanding star schema design in a practical context.